Feedback: achieving stable motion

M10 • 07 Jul 2027 - 08 Aug 2027 • 75 hours

What you will learn this month

By the end of the month, you will be able to explain why a joint reaches its target angle, why increasing gain produces oscillations, and how IMU delay affects platform stability. The deliverable is a benchmark for comparing joint and body-tilt control methods. Prepare a model, a digital controller, measured transient and frequency responses, a description of limitations and fault-test results. For a future walking robot, this provides the basis for maintaining body attitude; for a hand, it supports precise finger motion without accumulating error when pressing against an object.

Inputs and scope

Use the geometry, actuator, limits and measurements from M5-M9. Encoder and IMU reading, timestamping, and command and state logging must work before you start. Review torque, angular velocity, inertia, derivatives and integrals. Be able to read a step-response plot and distinguish measurement time from delivery time. If any of these elements is not ready, use the first two hours to diagnose it and reduce optional plots.

Begin with one axis. Full balancing of a multi-legged robot, gait planning and optimal force distribution are not required here. For tilt, use a platform model with limited rotation or a secured test rig from earlier stages. An IMU alone does not provide a plant to control: explicitly identify which actuator action changes roll or pitch. If this mechanism is unavailable, test tilt control in a model, physically measure only the joint and IMU, and leave physical tilt validation open.

What counts as a result

Show the continuous model, its discrete implementation and real data separately in the report. Give units for every coefficient and a source for every limit. The minimum final demonstration includes a joint-reference step, rejection of a repeatable disturbance, operation near saturation, tilt holding, artificial delay and loss of fresh data. Fix the criteria before the experiment, then report actual values without adjusting the acceptance limits.

The purpose of the month is to make control choices, beyond learning a list of controllers. Choose the control bandwidth, justify the loop frequency and show the conditions under which the solution stops working. A good result includes one clearly explained failure mode and the reason for excluding it from the operating region. This information will support the transition to state-space methods and estimation in M11.

Organising the 75 hours

Four work blocks

Block A, hours 1-20: feedback and joint PID. Block B, 21-40: frequency responses and noise. Block C, 41-60: digital implementation and loop timing. Block D, 61-75: body tilt and the final comparison experiment. Each of the first three blocks contains 10 hours of theory and analysis and 10 hours of practice; the fourth contains 5 hours of analysis and 10 hours of practice. Total: 35 + 40 = 75 hours. The work steps on pages 9-10 expand these same hours and require no additional time allocation.

Divide each ten-hour theory block into five two-hour sessions: 20 minutes reviewing the previous model, 45 minutes reading the relevant section, 40 minutes deriving a small example or examining a log, and 15 minutes recording the conclusion. Also divide the ten-hour Sunday practical session into five two-hour periods with breaks between them. Breaks are excluded from net study hours. Allocate the final five hours of analysis as 2 + 2 + 1.

Start with the question, then choose the tool

The main computational environment is Python, NumPy, SciPy and python-control; implement the control loop in C++ in the existing rig firmware. Begin each session with a testable question: how does the input affect the output, which state cannot be measured, and which data reveal its changes? Before running, write down the expected sign, scale or sequence of events. Compare the prediction with the result and examine at least one discrepancy. A plot supports a conclusion only when accompanied by an explanation of the observed relationship.

Maintain one repository with model, config, experiments, tests and reports directories. Every experiment needs a unique identifier, code version, configuration, parameter source, initial state, units and raw data. The analysis script reads a saved experiment and builds the report without manually correcting measurements. The final joint-and-attitude-control benchmark project combines comparisons of joint and body controllers; the individual exercises gradually form its test suite.

Time accounting and reducing scope

Calendar boundaries may coincide with an adjacent module. Keep rest days, holiday intervals and integration checks as specified in the overall calendar; place numbered steps in available time without counting hours twice. Do not add hours to this module. Assess equipment and library readiness at the outset rather than spending the practical session on an unplanned platform change.

If you fall behind, reduce the number of sinusoidal test frequencies and gain-scheduling variants; thoroughly check one body axis. Retain the joint model, a saturation experiment with anti-windup, frequency-response measurement, firmware timing assessment and the fault test for stale IMU data. You may investigate the second axis in a model if you state this in the report. Preserve the comparison of joint and tilt control; physical validation remains unfinished until there are experimental results from the rig.

Feedback, modelling and PID tuning

Study the relationship between torque and position

Examine the forward path command → actuator → mechanics → angle and the measurement feedback path. For the example joint, use $J\cdot\ddot{q} + b\cdot\dot{q} = \tau$: J in $kg\cdot m^2$, b in $N\cdot m\cdot s/rad$, τ in $N\cdot m$ and q in rad. Under torque control, the transfer function from τ to q is $1/(J\cdot s^2 + b\cdot s)$. If the actuator accepts only position, identify the actuator together with its internal closed loop as a single plant; a position command cannot replace torque in the model. Explain where the sensor, delay, saturation and disturbance appear in the diagram.

Poles determine natural modes; zeros change response shape. Stable poles of a continuous linear model have negative real parts; an integrator at zero needs separate reasoning, and simulation does not replace analysis. Study negative feedback $T = PC/(1 + PC)$, error $e = r - y$ and PID: proportional action responds to current error, integral action to accumulated error, and derivative action to rate of change. In a practical implementation, take the derivative of the measurement and filter it to reduce command kick when the reference changes.

Task 1. A joint model with three proportional-control responses

Inputs: estimated J and b or an explicitly labelled illustrative parameter set, a rest position and a bounded reference step. 1) Build the continuous model. 2) Choose three P values producing slow, acceptable and oscillatory responses. 3) Record angle, error and torque on the same time grid. 4) Explain changes in poles and curve shape. Deliverable: a computational notebook and a table of rise, peak and settling times. Check: the first motion has the correct sign relative to the command; doubling inertia must change the response. Error to investigate: reversed feedback sign, tested first in the model only.

Task 2. Integral action near the actuator limit

Add a constant external disturbance and the command limit from the rig specification. Compare P, PI and PID, then PID with anti-windup: for example, stop integrating when the command is already limited and the error demands more action in the same direction. Record the unsaturated command, limited command and integral state separately. Return the reference from an unreachable region to the admissible region and compare recovery times. Deliverable: a test and three plots. Check: the command never exceeds the declared limit; after returning, the integral must not hold the actuator in saturation beyond the expected recovery transient. Add feedforward as a separate calculation compensating for a known load, retaining feedback for the remaining error.

Frequency domain: bandwidth, margins and measurement

Study the meaning of a Bode plot

For sinusoidal excitation, frequency determines whether the joint can keep up with the command. Study magnitude and phase, logarithmic scales, 20·log10 of the amplitude ratio, and the conversion $\omega = 2\pi f$. Closed-loop bandwidth and open-loop crossover frequency are different quantities. Gain and phase margins are calculated from the loop transfer $L = PC$ under the chosen negative-feedback convention. A closed-loop response plot alone cannot establish a stability margin without knowing the loop structure.

Interpret $S = 1/(1 + L)$ and $T = L/(1 + L)$ as descriptions of disturbance rejection and signal and noise transmission at the chosen point in the diagram. State where the disturbance enters before comparing curves. Higher gain improves regulation over some low frequencies, but can amplify measurement noise and bring unwanted dynamics closer. A sensor filter changes phase as well as noise; include its delay in the loop analysis.

Task 3. Model margins and frequency response

Take the model from task 1 and the working PID. 1) Construct L and T and calculate margins with margin. 2) Check phase-argument units: phase data supplied to margin are in degrees, whereas frequency_response returns radians. 3) Introduce a known delay and rebuild the response. 4) Calculate step responses for the original and degraded cases. Deliverable: one sheet containing the Bode plot, poles and transient response. Check: a conclusion of reduced stability must agree with the time-domain experiment. If there is no crossover, explain the meaning of an infinite or undefined margin without treating it as a guarantee.

Task 4. A sinusoidal experiment on the rig

At small amplitude, apply 5-8 frequencies within the admissible range. At each frequency, wait for the transient to end, record at least several complete periods, and estimate amplitude and phase by fitting a sinusoid. Compare the reference-to-angle frequency response with model T. Deliverable: measured points with a range of variability and a configuration table. Check: repeat at half amplitude; a strong change in the ratio indicates nonlinearity or insufficient signal-to-noise ratio. Stop the experiment at the predefined limits if saturation, noticeable heating or excessive travel occurs. If margins are unconfirmed, describe tolerance to delay and load changes using the experimental series.

Digital control: discretisation and real time

Study system behaviour between samples

A digital controller reads a sensor at intervals Ts, calculates a command and holds it until the next update. Study the zero-order hold, conversion from continuous to discrete systems, and the ZOH and Tustin methods. A discrete linear model requires poles inside the unit circle for stability. Euler's method is useful for understanding, but an excessively large step creates numerical instability itself. Compare cont2discrete with a manual update of a simple first-order model.

Describe the loop period, computation time, measurement delay and data age separately. Jitter is variation in interval duration; quantisation is the finite resolution of a measurement or command. "A frequency of 200 Hz" without the distribution of actual intervals does not characterise operation. The Nyquist condition relates sampling frequency to recoverable signal bandwidth, but is not sufficient for choosing a control frequency: phase margins, filters, computation and the actuator need additional headroom, supported by modelling and experiment.

Task 5. One algorithm at several frequencies

Take one controller designed for the continuous model and discretise it for three Ts values selected relative to the desired bandwidth. Use the same physical plant, duration and reference amplitude. Introduce delays of 0, 1 and several samples. Deliverable: a Ts × delay experiment table, poles, RMSE and maximum command. Check: Ts is stored in the configuration, and integral and derivative actions scale with time. Deliberately retain old coefficients after changing Ts, show the error and add a test that detects it. Do not claim that example values apply to every actuator.

Task 6. Measure the embedded loop

In the existing firmware, timestamp the start of reading, end of calculation and command transmission using a monotonic clock; where possible, check the period with GPIO and a logic analyser. Run a quiet mode, intensive logging and an artificial delay in one handler. Save distributions of computation time, intervals and measurement age: median, p95, p99 and maximum, together with sample counts. Check: logging does not perform a blocking write in the critical section; a stale measurement does not become fresh through late delivery. Deliverable: a timing-budget report. Explicitly state the sensor polling, controller and logging frequencies; mark a repeated previous value rather than counting it as a new measurement.

Roll and pitch: controlling body tilt

Study the two-loop structure

The outer tilt controller receives roll and pitch and generates a bounded reference for the actuator’s inner loop. Check one axis first, then the other. The inner loop must track the outer-loop reference within the chosen range; tuning two fast loops independently without analysing their interaction can produce oscillations. Derive the relationship positive command change → tilt change from the geometry and verify it with a small test input. For a stationary model, distinguish tilt from linear acceleration, which distorts the accelerometer-based estimate.

Gain scheduling is specified in advance, for example as a function of position or load. It requires checks of boundaries and transitions, rather than arbitrary increases in P for large error. Limit the target angle, reference rate of change, inner-loop command and admissible IMU data age separately. Recovery after a fault must be an explicit state; the return of packets need not automatically resume motion.

Task 7. Holding a tilt angle

Initial conditions: a model or secured platform with a known axis, an IMU and a working inner controller. 1) Check zero and axis directions. 2) Command a small roll with zero pitch. 3) Repeat for pitch. 4) Introduce a repeatable disturbance pulse. 5) Compare controller-off and controller-on results under the same input. Deliverable: curves for both axes and commands. Check: calculate RMSE over the same window after the disturbance, peak error as the maximum absolute error, and settling time from entry into and continued residence within a predefined band. For a reference near zero, specify an absolute band in radians rather than a percentage of zero.

Task 8. Surface tilt, limits and recovery

Save two base orientations and two admissible loads; run every combination with fixed gains, then with two gain sets selected by operating point. Introduce a transition between gain sets and an IMU dropout. Deliverable: a table of stable and unconfirmed operating conditions, detection time and state sequence. Check: output continuity when gains change and correct handling of the integral; during a dropout, new motion commands stop within the time allocated for software fault detection. The physical stopping method depends on the mechanics: zero torque does not always hold a platform. Use the predefined behaviour of the secured rig and describe separately what a standing robot would require.

For the hand, repeat only the reasoning: object contact obstructs the commanded finger motion, the internal actuator reaches its limit and the integral accumulates error. Describe which implemented limits help and why a position PID alone is insufficient to control contact force.

Specification for the final benchmark

Defining a reproducible experiment

One scenario specifies the plant identifier, model version, initial angle, initial velocity, step or sinusoidal reference, disturbance, limits, Ts, filter and duration. The output log must contain measurement and control times, reference, measured joint angle or body tilt, estimated velocity, error, P, I and D terms, commands before and after limiting, saturation and deadline-miss flags, and state-machine state. Record the command in the units actually used and separate any conversion to torque.

For the joint, compare at least P, PID and PID with anti-windup. For the body, compare disabled stabilisation with the working outer loop, then one variant with increased delay. Every variant receives the same reference sequence and initial state. Save the scenario and configurations before comparison; do not choose different convenient segments for each algorithm. On the real rig, repeat the key scenario at least three times to distinguish a trend from a single successful run.

Calculating and interpreting metrics

RMSE = sqrt(mean(e^2)) has the units of the measured angle. Peak error = max(abs(e)); define percentage overshoot relative to a non-zero step amplitude. Calculate settling time as the time after which the signal remains inside the chosen band until the end of the window; if the window is too short, record “did not settle during the experiment”. The saturation fraction is the number of samples with a limited command divided by the number of commands. For uneven sampling, also calculate the fraction of time, since counting rows can distort the result.

Plots: reference and output; error and settling band; command and limits; data age; model Bode plot and measured points. Separate calibration and validation data: do not assess model suitability solely on the experiment used to fit J, b and delay. Use one validation experiment with a different admissible load or excitation frequency.

Deliverable-package requirements

A launch command from a clean working directory must build the baseline report and run the limit checks. The README explains the plant, wiring and mounting for physical experiments, parameters, limits and stopping sequence. Include a log of failed scenarios and the reasons for excluding them. State the basis for every conclusion: analytical linear model, nonlinear model, rig experiment or a hypothesis awaiting confirmation. This will allow M11 to compare LQR with the real PID using the same criteria.

Tests that reveal controller errors

Required fault scenarios

Scenario F1: reverse the encoder sign in the model. Expected: error grows rapidly and the software monitor detects departure from the permitted region; do not run this experiment physically. Save a short log up to stopping and demonstrate that the event is caused by the feedback sign, rather than random noise. The corrected configuration must pass the same test.

Scenario F2: an unreachable step followed by a return. Expected: the command is limited, the integral state is controlled, and recovery is faster than without anti-windup in the chosen test. Do not claim universal superiority from one curve; compare the specific recovery time and peak after the limit is removed.

Scenario F3: measurement delay and a stale packet. Separate transport delay from the loss of new measurements. Expected: the monitor uses the age of the original measurement, while frequency analysis and the time-domain model show degradation as delay increases. The last late packet must not reset the freshness timer without checking its timestamp.

Scenario F4: encoder noise and coarse quantisation. Expected: their effect on the D term and command is visible; a filter reduces some high-frequency noise, but its phase lag is represented in the model. Compare spectra and command RMS at a stationary reference, keeping this result separate from step-tracking accuracy.

Scenario F5: increased inertia or load. Choose a model change and a separate admissible rig load. Expected: metric changes that can be related to the model; the operating region is limited to measured cases. Do not mark untested masses and configurations as passed in the table.

Scenario F6: occasional long computations. Introduce one computation deadline miss every N cycles and batched logging. Expected: the event is explicitly marked, data losses are counted, and the response to lateness is checked. Compare p99 and the maximum; average duration must not hide an occasional overrun.

Analysing a failed experiment

First reproduce the failure using the saved configuration. Check units, signs and the time scale, then limiters, and only then gains. Change one factor per experiment. If the model is stable but the rig oscillates, check backlash, the actuator’s internal controller, the IMU filter, communication delay and frequency mismatches. Every unexplained effect becomes a separate task for additional measurements; reducing P without understanding the cause is a temporary measure.

Hours 1-40: foundations and frequency checks

Hours 1-20: joint and feedback

Theory 1-10: five 2-hour sessions. 1-2: examine the actuator interface and units; 3-4: derive the model and poles; 5-6: study the step response; 7-8: work through PID, feedforward and saturation; 9-10: design the sign and anti-windup checks. In each session, perform a small calculation using numbers from your configuration so that theory immediately reveals missing parameters.

Practice 11-20: 11-12: build the model and log; 13-14: complete task 1; 15-16: add integral and derivative actions and limiting; 17-18: complete task 2; 19-20: transfer one validated operating condition to the secured rig and save a validation recording. This is one 10-hour practical session in five periods; mounting preparation and the stopping check are included.

Before proceeding, check that you have a model with dimensions, three operating responses, a limited command and a return-from-saturation experiment. Be able to explain why an actuator with internal position control cannot be treated as an ideal torque source.

Hours 21-40: frequency and real data

Theory 21-30: 21-22: sinusoid, magnitude and phase; 23-24: Bode plots and crossovers; 25-26: gain and phase margins; 27-28: sensitivity and noise; 29-30: plan a frequency experiment with bounded amplitude. For one frequency, calculate the amplitude ratio and phase shift by hand, then compare with the library.

Practice 31-40: 31-32: complete task 3; 33-34: configure sinusoidal-response logging; 35-36: complete task 4 on the available rig; 37-38: repeat the end frequencies and half amplitude; 39-40: overlay the points and model, and document validity limits. If data are unusable, retain fewer frequencies with good repeated recordings rather than drawing a smooth curve from noise.

Before proceeding, verify that you know exactly what was measured: closed-loop T or loop transfer L. Magnitude and phase must have the correct units, and conclusions about margins must identify their basis.

Hours 41-75: the digital loop and body tilt

Hours 41-60: discretisation and the timing budget

Theory 41-50: 41-42: ZOH and the discrete model; 43-44: poles inside the unit circle; 45-46: Ts and PID coefficients; 47-48: delay, jitter and quantisation; 49-50: separating IMU, controller and logging frequencies. Draw a timeline for one measurement from physical sampling to command output and label every delay.

Practice 51-60: 51-52: complete task 5 for three Ts values; 53-54: add delays and the incorrect-coefficient test; 55-56: add firmware timestamps; 57-58: complete task 6 under load; 59-60: plot the distributions and reproduce a computation deadline miss. Do not spend these hours optimising before measuring the dominant source of delay.

Before proceeding, check that you have a timing distribution, each component’s original frequency, a stale-data test and an explanation of differences between the model and embedded implementation.

Hours 61-75: stabilisation and final validation

Analysis 61-65: 61-62: outer- and inner-loop diagram, actuation sign and limits; 63-64: RMSE, peak and settling criteria, and gain scheduling; 65: compile the final experiment table. These are the last 5 hours of theory and analysis, split as 2 + 2 + 1; do not add new chapters on multi-legged dynamics.

Practice 66-75: 66-67: complete task 7 for one axis; 68-69: repeat for the second and introduce a disturbance; 70-71: complete task 8 and the IMU-loss test; 72-73: repeat key scenarios and compile final metrics; 74-75: reproduce the package from a clean directory, write conclusions and list outstanding physical checks. If no platform is available, explicitly retain the status “tilt control validated in a model”.

Before proceeding, check that the comparison of joint and body control is complete and that command limits and fault behaviour have been tested. The next module receives a frozen baseline model and PID, rather than a set of unrelated demonstrations.

Acceptance and handover to M11

Practical defence of the result

In 10-15 minutes, show the loop diagram, one stable and one rejected operating condition, a frequency response and a fault recording. Explain how the model relates to the real actuator interface. Then change only delay or inertia and predict the direction of the response change in advance. A reviewer must be able to recalculate metrics from the raw recording and reach the same conclusion about whether the criteria were met.

Required evidence of readiness: positive axis directions are checked; settings and units are fixed; the declared operating region is bounded; saturation and loss of fresh data are tested; measured timing costs are documented. Support stability with margins from a correct loop model or experimental evidence over declared delays and loads. Do not describe the latter as a rigorous proof of stability for every possible operating condition.

Self-check questions

1) How does a pole differ from a zero in the joint example? 2) Why can increasing P both speed up and degrade the response? 3) What does the integral do when the physical command is already at its limit? 4) Why does differentiating the measurement differ from differentiating the error at a reference step? 5) Which transfer is used to calculate phase margin? 6) Why does the Nyquist frequency not determine a sufficient control frequency?

7) How can you distinguish sensor noise from loop oscillations? 8) What does a filter change besides noise amplitude? 9) How do you specify settling time for zero target tilt? 10) Why does average loop time not replace p99 and the maximum? 11) How can an old packet mistakenly appear fresh? 12) Why are zero torque and safe body holding not always equivalent? Support your answers with your own plots, as well as definitions.

What to save and which claims remain unconfirmed

Pass the model, baseline PID, discretisation parameters, disturbance scenarios, noise and delay configuration, and common metrics script to M11. They form the fixed basis for comparison with LQR and an estimator. In a separate table, list unmodelled effects: backlash, dry friction, torque-speed dependence, structural compliance and incorrect tilt estimation during acceleration.

If time is limited, retain one thoroughly tested body axis and a model demonstration of the second axis with explicit status. Reduce frequency count and presentation work, but retain closed-loop response measurement, saturation and fault scenarios. The module is complete only within the checks actually performed; any deferred physical work must appear in the subsequent integration plan.

Assigned reading: a focused reading route

Books and lecture notes may be in English; keep the plan, definitions and your own explanations in English. Read only the specified sections within the existing 75 hours.

Core reading and one reference route

1. Core: K. J. Åström, R. Murray, Feedback Systems, 2nd edition. Chapters Transfer Functions, Frequency Domain Analysis and PID Control; within the last chapter, Basic Control Functions, Integrator Windup and Implementation. For tasks 1-6, derive the loop equations, examine the effect of delay and implement a discrete PID.

2. Core worked example: B. Messner, D. Tilbury and the CTMS team, Control Tutorials for MATLAB and Simulink. The page Introduction: PID Controller Design, followed by DC Motor Speed: PID Controller Design. For tasks 1-2, reproduce the P, PI and PID comparison on your own model in Python.

3. Reference: J. Doyle, B. Francis, A. Tannenbaum, Feedback Control Theory. §3.1 Basic Feedback Loop and §3.4 Performance; §4.1 Plant Uncertainty - the introductory formulation only. For tasks 3-4, explain the trade-off between error, noise and model inaccuracy.

Documentation for specific practical work

Python Control step_info - transient-response criteria; margin - phase units and loop margins. Before plotting, check units: rad and deg, Hz and rad·s^{-1}.

SciPy cont2discrete - zoh, bilinear and euler for task 5. For the spectral analysis in F4, use the familiar scipy.signal.welch with explicit fs and scaling.

After reading

For each section, produce one plot and a short explanation using your own axis. Do not transfer coefficients from an illustrative example to the rig. MATLAB/Simulink are not mandatory purchases: translating the mathematical model into Python preserves the exercise’s purpose. Investigate delay in the model first, then repeat an admissible experiment on the rig.

Key terms and definitions

Plant. A physical system or its model whose output changes under a control input and external disturbances.

Control error. The difference between the reference and measured output, in the same units and a consistent coordinate frame.

Proportional action (P). The part of the command equal to current error multiplied by a gain.

Integral action (I). The part of the command that depends on error accumulated over time; it helps remove constant offset under admissible operating conditions.

Derivative action (D). The part of the command responding to the rate of change of error or measurement; it usually requires measurement-noise filtering.

Feedforward. A command calculated from the reference and a model of the expected input, without waiting for measured error to appear.

Transfer function. The ratio of the Laplace transforms of output and input for a linear time-invariant system under zero initial conditions.

Pole. A singularity of a transfer function associated with a natural dynamic mode; internal stability is checked with reference to the realisation.

Frequency response. The complex ratio of output and input amplitudes of a linear system under steady-state sinusoidal excitation at a specified frequency.

Bode plot. Plots of frequency-response magnitude and phase against a logarithmic frequency axis, with explicitly stated units.

Phase margin. The additional phase lag to the critical value at a selected unity-gain frequency of the open loop.

Gain margin. The permissible gain multiplier before the critical condition at a selected open-loop phase crossover.

Bandwidth. The frequency range over which a signal is reproduced acceptably; the boundary criterion must be declared for the selected channel.

Overshoot. The amount by which the output exceeds the final step level; a relative measure is normalised by the non-zero amplitude of that step.

Settling time. The time after which error remains within the declared band until observation ends for the given scenario.

Sampling period. The interval T_s between scheduled digital-controller updates; actual intervals are also measured using timestamps.

Zero-order hold. Keeping the last command constant until the next output update.

Jitter. Variation in actual execution times or update intervals relative to the chosen digital-loop schedule.

Gain scheduling. A predefined choice of parameters based on operating point, with validated transitions between modes.

Check the relationships: for unity negative feedback, $L=PC$, $T=L/(1+L)$, $S=1/(1+L)$. Frequency $\omega=2\pi f$.

Equipment and software preparation

USE EXISTING EQUIPMENT

MacBook Pro M4, the validated M7 rig, ESP32-S3, IMU and 8-channel logic analyser from M6, and the power supply and multimeter from M4. Use the tilt geometry and actuation axis from M8. Task 7 permits a model-based option; you do not need to buy a physical two-axis platform for this module.

PURCHASES: NO MANDATORY ADDITIONS

Optional: 1 oscilloscope only if current, ripple or signal edges cannot be assessed with existing equipment. Minimum: 2 channels, triggering, recording export and probes with suitable ratings. Choose bandwidth and sample rate for the fastest signal under investigation, rather than the mechanical-motion frequency.

Before buying, check data export on macOS ARM and input types. H-bridge potentials require a suitable differential measurement path; do not connect an ordinary grounded probe across two switching terminals. Selection guide: Tektronix: XYZs of Oscilloscopes.

INSTALL NOW: BEFORE HOURS 1-20

Add the **control** package (Python Control Systems Library) to the pinned macOS ARM environment, following Introduction → Installation. Use the existing NumPy, SciPy, Matplotlib and pytest installations. Slycot is optional for this module’s SISO experiments; retain the M6 ESP-IDF firmware environment.

INSTALLATION CHECK

Import control, print its version, create a first-order model and plot its step response. Compare the specified time constant with the plot; calculate step_info, the frequency response and margin for your loop. Obtain a discrete model through cont2discrete and compare it with the continuous model at small Ts.

Build the previous firmware on the microcontroller; measure the actual GPIO period with the logic analyser and compare it with timestamps. Save p95 and p99, the maximum, data age and a deadline-miss scenario in the log. Separately check stopping and recovery without spontaneous motion resumption.

You may repeat the calculations in a separate Ubuntu 24.04 ARM64 environment; test actual USB passthrough and virtual-machine graphics. Preparation is included in the 75 hours; order the instrument in advance if it proves necessary.

Motion fundamentals: holding and tracking

Required: why a stable position is insufficient

Holding specifies a constant angle qd; tracking specifies qd(t), velocity and acceleration. A leg joint must change angle on time during foot swing. Small error at the end of motion is insufficient: lag along the path changes foot position. Before calculating, explain the negative-feedback sign and the distinction between angle, angular velocity and torque.

Joint model and error equation

The base is fixed; positive q and τ are chosen along the same axis. q is in rad, and primes denote time derivatives: q′ in rad/s, q″ in rad/s². The link moves without gravitational torque; J is inertia about the joint in kg·m², and b is viscous resistance in N·m·s/rad. For now, assume no saturation, delay or model error:

J q″ + b q′ = τ; e = qd − q. Choose **τ = J qd″ + b qd′ + Kp e + Kd e′**. The first two terms provide acceleration and overcome resistance in advance; PD corrects deviation. Substituting q = qd − e into the model gives **J e″ + (b + Kd)e′ + Kp e = 0**. This is error dynamics, not a separate physical joint.

Let J = 0.02 kg·m², b = 0.04 N·m·s/rad, Kp = 2 N·m/rad and Kd = 0.36 N·m·s/rad. Then e″ + 20e′ + 100e = 0; both poles equal $-10\ s^{-1}$. Error decays with critical damping in the ideal continuous model. This conclusion alone does not confirm the stability of a digital actuator with limits.

Calculation at a selected instant

At the selected instant, qd = 0.30 rad and q = 0.20 rad; qd′ = q′ = 0.50 rad/s and qd″ = 1 rad/s². Thus e = 0.10 rad and e′ = 0; feedforward is 0.02·1 + 0.04·0.50 = 0.04 N·m, feedback is 0.20 N·m, and total τ = 0.24 N·m. The model gives q″ = (0.24 − 0.04·0.50)/0.02 = 11 rad/s² and e″ = −10 rad/s²: the joint accelerates, reducing its lag behind the reference.

If feedforward is removed and qd moves at a constant velocity of 0.50 rad/s for a long time, steady-state e = b·qd′/Kp = 0.01 rad. The error creates the torque needed to overcome resistance. For a stationary target in this model, e tends to zero. Holding and tracking therefore require different test references.

Two experiments within the existing laboratory work

1. For a constant target, take e(0) = 0.10 rad and e′(0) = 0. Save an e plot and compare with e(t) = 0.10(1 + 10t)exp(−10t): at t = 0.40 s, expect 0.00916 rad. **2.** For a linearly changing reference, compare PD with and without feedforward; save qd, q, τ and the individual terms. Check: error without feedforward approaches 0.01 rad; mark saturated segments separately.

Time: 1 theory hour from hours 7-8 (PID and feedforward), and 2 practice hours from 15-16 (integral and derivative actions, limits); this analysis and these plots belong to existing tasks 1-2, within the 75 hours. The numbers apply to the model. Do not send torque τ to a position servo: its internal loop is studied separately. Full multi-joint computed-torque control remains outside the mandatory minimum.

Read selectively: Lynch, Park, Modern Robotics, §11.4, Part 1: PD and error dynamics; §11.4, Part 3: model-based feedforward.